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Abstract

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[Dedication]

Definitions and abbreviations

List of Acronyms

Symbol	Description
DNS	Numerical solution resolving all relevant flow scales
NASA	United States civilian space agency

Greek Symbols

Symbol	Description	Units
γ	Ratio of specific heats	1
μ	Dynamic viscosity	$N\ s\ m^{-2}$
ρ	Fluid density	$kg\ m^{-3}$

Roman Symbols

Symbol	Description	Units
T	Temperature	K
p	Pressure	Pa
u	Velocity	$m\,s^{-1}$

Superscripts

Symbol	Description
*	Dimensional quantity

Subscripts

Symbol	Description
∞	Free-stream condition
s	Species index

Dimensionless Numbers

Symbol	Description	Definition
Pr	Prandtl number	$\mu c_p / \kappa$
Re	Reynolds number	$\rho u L / \mu$

Appendix A

Appendix A

Appendix B

Appendix B

References